

JYOTI KUMARI

Ranchi, Jharkhand

📞 +91-8797075142 ✉ jyotika918273@gmail.com [in linkedin.com](https://www.linkedin.com/in/jyotikakumari) [github.com](https://github.com/jyotikakumari) [Portfolio](#)

Education

Birla Institute of Technology, Mesra Ranchi

Sep. 2023 – May 2027

Bachelor of Technology, Electronics and Communication Engineering

GPA: 8.5

Delhi Public School, Dhurwa Ranchi

May. 2020 – May 2022

AISSCE (Class XII)

Percentage: 91%

St. Anne's Girls' School, Ranchi

May 2010 – May 2020

Secondary Education (Class X)

Percentage: 95%

Projects

Musix | *MERN, Tailwind CSS, Socket.io, JWT*

[Live Demo](#) | [GitHub](#)

- Built a full-stack Spotify-inspired music streaming platform serving dynamic playlists, **real-time chat**, and **seamless audio playback**, improving user engagement through an interactive multi-module experience.
- Developed an **admin management dashboard** handling songs, albums, artists, and users with optimized **CRUD operations**, reducing manual content management effort by automating catalog updates and monitoring.
- Integrated **real-time messaging**, **live listening activity tracking**, and responsive dark-themed UI delivering a scalable and modern music streaming experience across devices.

BitConnect | *MERN Stack*

[Live Demo](#) | [GitHub](#)

- Developed **RESTful APIs** for secure user authentication, post-creation, and community engagement features using **Express.js** and **MongoDB**. Reduced the average API response time by **40%** through **efficient database indexing**.
- Engineered a high-traffic event portal featuring **Real-time WebSocket notifications/RSVP tracking/Social media integration**, effectively centralizing activity for **15+** student organizations and improving user engagement.
- Strengthened platform security by implementing **JWT (JSON Web Tokens)** and **bcrypt-based** authentication protocols alongside a granular **Role-Based Access Control (RBAC)** system for Admins, Clubs, and Students.

Article to podcast Agent | *Python, React.js*

[Live Demo](#) | [GitHub](#)

- Architected a sophisticated end-to-end **AI pipeline utilizing Agno agents and OpenAI GPT-4o** to transform static web articles into immersive audio experiences through **Firecrawl scraping** and **ElevenLabs API** for voice synthesis.
- Developed a **centralized Streamlit dashboard** that seamlessly integrates asynchronous audio processing pipelines with real-time playback capabilities; engineered a resilient error-handling layer to dynamically parse diverse web schemas.
- Refined autonomous agentic workflows and tool-calling logic to **enforce deterministic chaining**, which improved the reliability of the "Scrape-to-Audio" pipeline by **30%** during high-concurrency testing under simulated production loads.

Skills Summary

Languages: C++, C, Python, Java, HTML/CSS, JavaScript, SQL

Web Technologies: React.js, Next.js, TypeScript, Node.js, Express.js, RESTful APIs

Databases: MySQL, PostgreSQL, MongoDB

Developer Tools: Git, GitHub, Postman, Linux, Docker

Cloud : AWS (EC2, S3, CloudWatch – basics)

ML & Data: Scikit-Learn, Pandas, NumPy, PyTorch, TensorFlow, LangChain

Core Concepts: DSA, OOPs, DBMS, Operating Systems, Computer Networks, System Design

Achievements

- Participated in the **EY Hackathon** built a **loanease web-based AI-driven** conversational assistant designed to automate the personal loan acquisition process by solving the problem proposed by **TATA capitals**.
- Maintained a competitive **LeetCode rating of 1710 (Top 10%)** and a 2-star rating on CodeChef, demonstrating consistent problem-solving skills across 300+ challenges in **Dynamic Programming, Graphs, and Trees**.
- Actively contributed to the competitive programming community through frequent participation in **Codeforces** and **CodeChef** global contests, refining algorithmic efficiency under strict time constraints.

Positions of Responsibility

Institute of Engineering and Technology(IET)

Sep. 2024 – Present

Executive Body Member

BIT Mesra

- Organized and facilitated **3+ coding competitions** with **200+ participants**, improving the campus coding culture.
- Managed executive board of 5 members and ran weekly meetings to oversee progress in essential parts of the club.
- Mentored junior developers**, providing guidance on coding practices and project execution